

Experiment 10**Inventory Management Visualization**

Aim: To create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

Procedure**Step 1: Set Up Your HTML File (index.html)**

This file sets up the structure of the dashboard and links the necessary Chart.js library.

HTML

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <title>Inventory Dashboard</title>
6
7      <link rel="stylesheet" href="style.css">
8
9      <!-- Google Font -->
10     <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;600&display=swap" rel="stylesheet">
11
12     <!-- Chart.js -->
13     <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
14 </head>
15 <body>
16
17     <div class="container">
18         <h1>Inventory Dashboard</h1>
19         <p class="subtitle">Simple and clean stock visualization</p>
20
21         <div class="charts">
22
23             <div class="card">
24                 <h2>Distribution</h2>
25                 <canvas id="pieChart"></canvas>
26             </div>
27
28             <div class="card">
29                 <h2>Stock Levels</h2>
30                 <canvas id="barChart"></canvas>
31             </div>
32
33         </div>
34     </div>
35
36     <script src="script.js"></script>
37 </body>
38 </html>
```

Step 2: Create the CSS File (style.css)

The CSS ensures the interface is clean, professional, and follows the "Inventory Dashboard" design.

```
1  body {
2      margin: 0;
3      font-family: 'Poppins', sans-serif;
4      background-color: #f7f9fc;
5      color: #2c3e50;
6  }
7
8  /* Container */
9  .container {
10     text-align: center;
11     padding: 40px 20px;
12 }
13
14 /* Heading */
15 h1 {
16     font-size: 2.2rem;
17     margin-bottom: 5px;
18 }
19
20 .subtitle {
21     color: #7f8c8d;
22     margin-bottom: 30px;
23 }
24
25 /* Layout */
26 .charts {
27     display: flex;
28     justify-content: center;
29     gap: 25px;
30     flex-wrap: wrap;
31 }
32
33 /* Card */
34 .card {
35     background: #ffffff;
36     border-radius: 15px;
37     padding: 20px;
38     width: 380px;
39     box-shadow: 0 6px 20px rgba(0, 0, 0, 0.08);
40     transition: 0.2s ease;
41 }
42
43 .card:hover {
44     transform: translateY(-5px);
45     box-shadow: 0 10px 25px rgba(0, 0, 0, 0.12);
46 }
47
48 h2 {
49     font-weight: 600;
50 }
51
52 /* Canvas */
53 canvas {
54     margin-top: 15px;
55 }
```

Step 3: Create the JavaScript File (script.js)

This script defines the inventory data and initializes both the Pie and Bar charts.

JavaScript

```

1  const labels = ["Electronics", "Clothing", "Home Appliances", "Books", "Toys"];
2  const dataValues = [200, 150, 100, 80, 50];
3
4  const colors = [
5    "#a8dadc",
6    "#ffd6a5",
7    "#bdb4db",
8    "#bde0fe",
9    "#ffa5cc"
10 ];
11
12 const textColor = "#2c3e50";
13
14 // PIE CHART
15 new Chart(document.getElementById("pieChart"), {
16   type: "pie",
17   data: {
18     labels: labels,
19     datasets: [{
20       data: dataValues,
21       backgroundColor: colors
22     }]
23   },
24   options: {
25     plugins: {
26       legend: {
27         labels: {
28           color: textColor
29         }
30       }
31     }
32   }
33 });

```

```

34
35
36 // BAR CHART
37 new Chart(document.getElementById("barChart"), {
38   type: "bar",
39   data: {
40     labels: labels,
41     datasets: [{
42       label: "Items in Stock",
43       data: dataValues,
44       backgroundColor: colors
45     }]
46   },
47   options: {
48     scales: {
49       x: {
50         ticks: { color: textColor }
51       },
52       y: {
53         beginAtZero: true,
54         ticks: { color: textColor }
55       }
56     },
57     plugins: {
58       legend: {
59         labels: {
60           color: textColor
61         }
62       }
63     }
64   }
65 });

```

Output

The resulting dashboard displays two distinct visualizations:

- **Pie Chart:** Titled "Inventory Distribution," showing the percentage of stock across five categories.
- **Bar Chart:** Titled "Items in Stock by Category," plotting the exact quantity of items (ranging from 0 to 200) for each category.
- **Visual Layout:** The dashboard features a clean interface with a header and organized cards for each chart

